

Jared Schultz

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EDUCATION

University of Minnesota, College of Science and Engineering

Minneapolis, MN

Masters of Science, Robotics, Minor in Mechanical Engineering

9/2024 – 5/2026

GPA: 4.0

University of Minnesota, College of Science and Engineering

Minneapolis, MN

Bachelor of Science, Computer Engineering

9/2020 – 5/2024

GPA: 3.52

TECHNICAL SKILLS

Robotics: Computer Vision, Machine Learning, AI, Path Planning, Kalman Filtering, Control Theory, Torque Response

Programming: C, Java, Python, ROS, Bash, Verilog

Mechanical: SolidWorks, RoboDK, 3d Printing (SLA and FDM), Machine shop trained 100+ hours

Electrical: DC/AC circuit analysis, Altium Designer

Tools and Platforms: Unix, MATLAB, Simulink, Vivado, Git, Jira, Confluence

JOB EXPERIENCE

University of Minnesota and Medtronic

Minneapolis, MN

Lead Mechatronic Designer for TRUST Surgical Stroke Robot

6/2025 – Present

- Develop a comprehensive test suite to characterize prototype performance, accelerate wear, and identify failure modes to promote device longevity.
- Design, manufacture, and test iterative robot prototypes, using results to verify and inform design revisions.
- Design and implement a motor control architecture enabling precise control of the guidewire propulsion mechanism.

General Dynamics Mission Systems

Bloomington, Minnesota

Software Engineering Intern

5/2022 – 8/2022

- Translated the existing test environment from Bash into Python to improve readability and package support.
- Anticipated future requirements and implemented unit and system tests into the active test environment.
- Contributed to professional documentation through Confluence, understood team workflow through Jira, and streamlined version control through GitLab.

PROJECT AND RESEARCH EXPERIENCE

Robotics Club projects (Robo Master, NASA Lunabotics, Robot in Three Days)

1/2021 – Present

- Work under a time crunch to design and manufacture a robot to fulfill a task as specified by the competition.
- Develop a computer vision system to detect, track, and predict motion in a dynamic camera-target environment
- Collaborate with my team to design, manufacture, and validate robotic subsystems to ensure reliable task performance.

Intermediate Robotics Project

9/2024 – 12/2024

- Used torque measurements to implement feedback and measure the effects of delay on manipulability
- Developed methods for control, teleoperation, and feedback for UR5 Robot over the University network

Robot Vision Final Project

9/2024 – 12/2024

- Designed and built from scratch an autonomous 6 DOF arm able to simulate industrial pick-and-place tasks
- Developed and debugged a custom position-based visual servoing algorithm to allow for dynamic object tracking.

Autonomous Quadrotor for Education

12/2023 – 5/2024

- Assisted a professor in exploratory research and completed the initial steps in creating a new class for autonomous aerial flight and tracking.
- Implemented signal processing and path planning algorithms in a pipeline to allow the real-time position of the drone to be used to follow a path autonomously.

RELEVANT CLASS EXPERIENCE

ME 8284 Intermediate Robotics

ME 5286 Robotics

ME 5281 Feedback Control Systems

EE 5271 Robot Vision

EE5231 Linear Control Systems

CSCI 8551 Intelligent Agents

CSCI 5801 Software Engineering 1

CSCI 5551 Introduction to Intelligent Robotic Design

CSCI 5521 Machine Learning Fundamentals

CSCI 5511 Artificial Intelligence